VeriSight: Master Codebase & Architecture Journal

Comprehensive File-by-File Technical Guide with Real Implementation Code, Design Rationales & Execution Mechanics

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Executive Codebase Overview: This journal is a comprehensive, code-level reference for the VeriSight platform. Unlike high-level marketing overviews, this document inspects the **exact Python and SystemVerilog code implementations** written across the verisight/ package. For every core file, it provides: (1) The exact directory path, (2) The actual code snippet written in that file, (3) **Why that code was written** (the specific engineering problem, failure mode, or hallucination risk it eliminates), and (4) **How it works** (the step-by-step execution flow during simulation triage, verification planning, or UVM synthesis).

1. User Experience, Wizard Launcher & Master Orchestration

■ File Path: verisight/wizard.py

System Role: Interactive Terminal Launcher & 2x2 Tri-Engine Gateway

```
Core Implementation Code:
  # Pure Spec-Driven Mode Handling in VeriSight Plan:
  if not rtl_path:
   # RTL is strictly optional: default is empty string so pressing Enter skips
   rtl_path = _prompt_path("RTL Source directory or file (.sv, .v)", required=False, default="")
  ...
  summary_table.add_row("RTL Source", str(rtl_path) if rtl_path else "[bold green]None (Spec-Driven
Planning)[/bold green]")
```

■ **Why We Wrote This Code:** Previously, when users did not provide an RTL file, the wizard defaulted to examples/rtl/alu.sv, which caused the AI to hallucinate ALU logic into non-ALU specifications (such as I2C or FIFO). We rewrote this to enforce clean, pure spec-driven execution where RTL is optional, with a 2x2 interactive card interface displaying silicon hero badges.

■■ **How It Works & Logic:** The wizard prompts the user for specifications and configuration parameters. When the user selects Engine 2 and hits Enter to skip RTL, it marks rtl_path as None and passes this flag downstream to the planner orchestrator, enabling true pre-RTL shift-left verification planning.

■ **Inter-Module Dataflow:** Dispatches execution to verisight/orchestrator.py, verisight/plan/planner_orchestrator.py, or verisight/agents/spec_driven_tb_agent.py.

■ File Path: verisight/orchestrator.py

System Role: Multi-Agent Cognitive Pipeline Orchestrator (Engine 1)

Core Implementation Code:

```
class VeriSightOrchestrator:
 def run_pipeline(self, rtl_path, log_path, spec_path=None, vcd_path=None):
 knowledge = self.agent1.execute(rtl=rtl_path, log=log_path, spec=spec_path, vcd=vcd_path)
 classification = self.agent2.execute(knowledge=knowledge)
 rtl_analysis = self.agent3.execute(knowledge=knowledge, classification=classification)
 fix_result = self.agent5.execute(knowledge=knowledge, rtl_analysis=rtl_analysis)
 report = self.agent4.execute(knowledge=knowledge, classification=classification, fix=fix_result)
 return report
```

■ **Why We Wrote This Code:** Prevents monolithic LLM failure. When a single prompt attempts log parsing, AST elaboration, classification, and code patching, models get confused and hallucinate non-existent hardware. We wrote this orchestrator to enforce strict separation of cognitive duties across 5 independent agents.

■■ **How It Works & Logic:** Executes a linear, strongly-typed pipeline where each agent consumes validated Pydantic artifacts from the previous agent. If any agent encounters an LLM timeout, the orchestrator triggers graceful offline fallback to local deterministic rule engines.

- ■ **Inter-Module Dataflow:** Sequences Agent 1 -> Agent 2 -> Agent 3 -> Agent 5 -> Agent 4.

■ File Path: verisight/config.py

System Role: Multi-Provider Configuration & Air-Gap Model Resolver

Core Implementation Code:

```
class VeriSightConfig(BaseModel):
 model_provider: str = Field(default="gemini") # gemini, claude, openai, ollama, vllm
 airgap_mode: bool = Field(default=False)
 local_endpoint: str = Field(default="http://localhost:11434")
 token_budget_per_run: int = Field(default=15000)
```

■ **Why We Wrote This Code:** Enforces strict semiconductor confidentiality and vendor independence. Chip companies cannot send unmasked RTL to external clouds. This configuration allows zero-code switching between commercial cloud APIs and 100% on-premise air-gapped models.

■■ **How It Works & Logic:** Loads .env variables, validates API keys with pre-flight probes, and routes agent requests to either cloud endpoints (Gemini/Claude) or local Ollama/vLLM endpoints (DeepSeek-Coder/Llama-3).

- ■ **Inter-Module Dataflow:** Loaded by base_agent.py and all tools.

2. The Specialized Multi-Agent Intelligence Fabric

■ File Path: verisight/agents/base_agent.py

System Role: Abstract Base Agent with Schema Enforcement & Backoff

```
Core Implementation Code:
  class BaseAgent:
   def reason(self, prompt: str, system_prompt: str, output_model: Type[BaseModel]) -> BaseModel:
   # Enforces structured output parsing and exponential backoff retry
   for attempt in range(self.max_retries):
   raw_response = self._call_provider(prompt, system_prompt)
   validated = self._validate_and_parse(raw_response, output_model)
   if validated:
   return validated
   return self._fallback_deterministic(output_model)
```

■ **Why We Wrote This Code:** Raw LLMs output conversational text that breaks downstream scripts. We wrote BaseAgent to guarantee that every single agent response strictly validates against a Pydantic schema, and that transient network glitches are automatically retried with exponential backoff.

■■ **How It Works & Logic:** Wraps Google GenAI, Anthropic, and OpenAI SDKs. Parses JSON from raw LLM responses, strips markdown wrappers, validates against the target Pydantic schema, and falls back to deterministic rule evaluations if retries are exhausted.

■ **Inter-Module Dataflow:** Inherited by all 6 agent classes in verisight/agents/.

■ File Path: verisight/agents/agent1_knowledge.py

System Role: Agent 1: Knowledge Engine & Multi-Modal Collateral Ingestion

```
Core Implementation Code:
  class KnowledgeAgent(BaseAgent):
   def extract_knowledge(self, rtl_path, log_path, spec_path, vcd_path):
   ast_data = self.pyslang_parser.parse(rtl_path)
   log_errors = self.log_parser.tokenize(log_path)
   vcd_slice = self.vcd_parser.slice_window(vcd_path, window=20)
   return UnifiedKnowledge(ast=ast_data, logs=log_errors, waveform=vcd_slice)
```

■ **Why We Wrote This Code:** Simulation failures involve disjoint files (logs, Verilog source, waveforms, specs). An engineer wastes hours correlating lines. Agent 1 bridges these media into a single coherent knowledge graph before any diagnosis begins.

■■ **How It Works & Logic:** Invokes pyslang_rtl.py to elaborate the AST, log_parser.py to extract fatal error signatures, and vcd_parser.py to perform temporal window slicing (+/- 20 cycles), returning UnifiedKnowledge.

■ **Inter-Module Dataflow:** Emits knowledge.json consumed by Agent 2.

■ File Path: verisight/agents/agent2_classifier.py

System Role: Agent 2: 9-Rule Decision Engine & Evidence Calibrator

```
Core Implementation Code:
  class RootCauseClassifierAgent(BaseAgent):
   def _evaluate_rules(self, knowledge: UnifiedKnowledge) -> Classification:
   # Rule 1: Scoreboard Mismatch (DUT output != Reference output)
   # Rule 2: X-Propagation (4-state logic uninitialized flop)
   # Rule 3: Protocol Handshake Violation (SVA assertion failure)
   ...
   classification = self._verify_and_calibrate(classification, knowledge)
   return classification
```

■ **Why We Wrote This Code:** Prevents ungrounded AI hallucination. We wrote this code to enforce a strict rule-based taxonomy: a failure cannot be declared an 'RTL Bug' unless physical evidence (AST lines, timestamps, waveform transitions) satisfies one of the 9 formal rules.

■■ **How It Works & Logic:** Executes a deterministic pre-analysis, sends evidence to the LLM with strict instructions, and then runs _verify_and_calibrate() to cross-verify the LLM's citations against actual source files, penalizing confidence if any hallucinated identifier is detected.

■ **Inter-Module Dataflow:** Emits summary.json consumed by Agents 3, 4, and 5.

■ File Path: verisight/agents/agent3_rtl_analyzer.py

System Role: Agent 3: RTL Root Cause Analyzer & Logic Cone Tracing

```
Core Implementation Code:
  class RTLAnalyzerAgent(BaseAgent):
   def analyze_rtl(self, knowledge: UnifiedKnowledge, summary: Classification):
   x_origins = self.xtracer.find_uninitialized_regs(knowledge.ast)
   cone_signals = self.traceweave.backward_cone(target_signal, cycle=summary.fail_cycle)
   cdc_hazards = self.extended_checks.check_cdc(knowledge.ast)
   fsm_deadlocks = self._check_fsm_deadlocks(knowledge.ast)
   return RTLAnalysisReport(x_origins=x_origins, cone=cone_signals, cdc=cdc_hazards)
```

■ **Why We Wrote This Code:** Identifies circuit-level defect mechanisms. Even after knowing a bug is in the RTL, isolating whether it is an uninitialized register, a clock-domain crossing race, or an FSM deadlock normally takes hours of waveform tracing.

■■ **How It Works & Logic:** Executes the static X-Tracer to find reset-less flops, walks backward through the AST fan-in cone using WaveTrace, checks for missing 2-FF CDC synchronizers, and inspects FSM transition graphs for deadlocks.

■ **Inter-Module Dataflow:** Emits rtl_analysis.json consumed by Agent 4 and Agent 5.

■ File Path: verisight/agents/agent4_reporter.py

System Role: Agent 4: Interactive Dashboard & Report Compiler

```
Core Implementation Code:
  class ReporterAgent(BaseAgent):
   def generate_reports(self, knowledge, classification, rtl_analysis, fix_result):
   html_dashboard = self._build_interactive_html(classification, rtl_analysis, fix_result)
   markdown_report = self._compile_markdown_summary(classification, rtl_analysis)
   json_error = ErrorSchema(root_cause=classification.reason, patch=fix_result.patch)
   return ReportArtifacts(html=html_dashboard, md=markdown_report, json=json_error)
```

■ **Why We Wrote This Code:** Transforms complex verification data into actionable executive and technical views. Eliminates manual reporting by providing synchronized waveform cursors and machine-readable JSON for CI/CD pipelines.

■■ **How It Works & Logic:** Aggregates diagnostic data from all agents, embeds sub-cycle waveform timelines via waveform_studio.py, formats unified diff views, and writes report.html, report.md, and error.json.

■ **Inter-Module Dataflow:** Final deliverable compiler; outputs to output/ directory.

■ File Path: verisight/agents/agent5_fix_generator.py

System Role: Agent 5: Minimal Surgical Git Patch Synthesizer & Sandbox

```
Core Implementation Code:
  class FixGeneratorAgent(BaseAgent):
   def generate_fix(self, rtl_analysis: RTLAnalysisReport):
   # Enforces minimal code region patch (lines starting with - and +)
   diff_text = self._synthesize_patch_diff(rtl_analysis)
   # Verify syntax with pyslang before testing
   if not self.sv_validator.validate_syntax(diff_text):
   return FixResult(available=False, reason="Syntax check failed")
   # Validate in isolated /tmp sandbox simulation
   passed = self.sim_runner.run_sandbox_regression(diff_text)
   return FixResult(available=passed, patch=diff_text)
```

■ **Why We Wrote This Code:** Prevents AI from generating broken or non-synthesizable code fixes. Many AI tools rewrite whole files, introducing new bugs. Agent 5 operates under a strict minimal-diff policy and tests every patch in a compilation sandbox before presenting it.

■■ **How It Works & Logic:** Identifies the exact buggy lines, creates a minimal unified git diff, validates syntax via pyslang, copies design files to a temporary /tmp sandbox, and re-executes simulation to ensure the failing assertion passes without regression.

■ **Inter-Module Dataflow:** Emits proposed.patch and updates fix_result consumed by Agent 4.

■ File Path: verisight/agents/spec_driven_tb_agent.py

System Role: Engine 3: 10-Stage Spec-Driven UVM 1.2 Synthesis Agent

```
Core Implementation Code:
  class SpecDrivenTBAgent(BaseAgent):
   def generate_testbench(self, spec_path: str, rtl_path: str, output_dir: str):
   # Stage 1-2: Spec Ingestion & Grounding (Spec is the functional source of truth)
   spec_truth = self._extract_spec_truth(spec_path)
   # Stage 3-4: Interface Extraction via pyslang AST
   ports = self.pyslang.extract_ports(rtl_path)
   # Stage 5-8: Synthesize 14-Component UVM VIP Suite with Independent Golden Ref Model
   suite = self.uvm_architect.synthesize(spec_truth=spec_truth, ports=ports)
   # Stage 9-10: Pre-compile with pyslang & run dynamic simulation sign-off
   self.sv_validator.validate_suite(suite)
   return SpecDrivenTBResult(suite=suite)
```

■ **Why We Wrote This Code:** Solves the 'AI testbench mirroring' flaw. When an AI reads buggy RTL to write a testbench, it copies the designer's bugs into the reference model, producing 100% false passes. This agent enforces the 'Spec-as-Truth' principle: the reference model is grounded strictly in the specification.

■■ **How It Works & Logic:** Executes a structured 10-stage pipeline: ingests specs, maps RTL interfaces via pyslang AST, synthesizes 14 VIP components (including golden reference model and scoreboard comparator), checks IEEE-1800 syntax, and verifies simulation.

■ **Inter-Module Dataflow:** Primary engine for Workflow 4 / Engine 3; uses verisight/uvm/uvm_architect.py.

3. Engine 2 Deep Dive: Scaled Verification Planning & Coverage

■ File Path: verisight/plan/planner_orchestrator.py

System Role: Verification Plan Master Orchestrator (Spec-Driven Planning)

Core Implementation Code:

```
  class PlannerOrchestrator:
   def generate_verification_plan(self, spec_path, rtl_path=None, coverage_path=None):
   # Pure Spec-Driven Mode: RTL is optional
   spec_text = read_file(spec_path)
   features = self.feature_extractor.extract_features(spec_text, rtl_path=rtl_path)
   tests = self.test_planner.generate_tests(features) # Generates 50+ scaled tests
   covergroups = self.coverage_architect.build_coverage_model(features) # 90+ CPs
   assertions = self.assertion_architect.generate_assertions(features) # 30+ SVA
   return VerificationPlanReport(tests=tests, coverage=covergroups, sva=assertions)
```

■ **Why We Wrote This Code:** Coordinates all planning modules into a cohesive flow. Enables pure spec-driven verification planning directly from PDF/Markdown datasheets before RTL coding even starts.

■■ **How It Works & Logic:** Sequences Phase 1 (Feature Extraction), Phase 2 (50+ Test Scenario Formulation), Phase 3 (Coverage Model Synthesis), and Phase 4 (SVA Assertion Synthesis). Generates verification_plan.json and verification_plan.md.

■ **Inter-Module Dataflow:** Coordinates feature_extractor.py, test_planner.py, coverage_architect.py, and assertion_architect.py.

■ File Path: verisight/plan/feature_extractor.py

System Role: Multi-Modal Specification Feature Decomposer

Core Implementation Code:

```
  class FeatureExtractor:
   def extract_features(self, spec_text: str, rtl_path: Optional[str] = None):
   # Decomposes spec into atomic, verifiable items (e.g. REQ_001, REQ_002)
   sections = self._split_sections(spec_text)
   features = []
   for sec in sections:
   features.extend(self._extract_atomic_requirements(sec))
   return features
```

■ **Why We Wrote This Code:** Eliminates ambiguity in specifications. Specifications are written in free-form English, making it hard to track what is verified. This code breaks specifications down into atomic, numbered functional requirements.

■■ **How It Works & Logic:** Uses prompt engineering to parse sections, register maps, and operational modes, outputting normalized RequirementItem objects with unique IDs, categories, and verification priorities.

■ **Inter-Module Dataflow:** Feeds extracted requirements to test_planner.py and coverage_architect.py.

■ File Path: verisight/plan/test_planner.py

System Role: 50+ Scaled Test Matrix Synthesizer (8 Functional Classes)

```
Core Implementation Code:
  class TestPlanner:
   CATEGORIES = [
   "directed", "constrained_random", "reset_initialization", "boundary_corner",
   "error_negative", "performance_backpressure", "concurrency_race", "protocol_specific"
   ]
   def generate_tests(self, features: List[RequirementItem]) -> List[TestCaseItem]:
   # Formulates at least 6-8 unique tests for each of the 8 categories -> 50+ tests
   tests = []
   for cat in self.CATEGORIES:
   tests.extend(self._synthesize_category_scenarios(features, category=cat))
   return tests
```

■ **Why We Wrote This Code:** Solves the 'shallow test plan' problem. Earlier tools generated only 10 to 12 generic tests, missing corner cases. We wrote this code to enforce an 8-category verification matrix guaranteeing at least 50 unique, scaled test scenarios.

■■ **How It Works & Logic:** Iterates across all 8 verification dimensions, prompting the LLM with category-specific constraints (e.g. inject bus parity errors for 'error_negative'; test FIFO simultaneous read/write for 'boundary_corner').

■ **Inter-Module Dataflow:** Outputs test_plan.json and test_plan.md.

■ File Path: verisight/plan/coverage_architect.py

System Role: Multi-Covergroup & Functional Coverage Model Synthesizer

```
Core Implementation Code:
  class CoverageArchitect:
   def build_coverage_model(self, features: List[RequirementItem]) -> CoveragePlan:
   # Synthesizes 90+ coverage points and 150+ bins
   cg_models = []
   for feat in features:
   cg = self._synthesize_covergroup(feat)
   cg.add_cross_coverage(feat.cross_signals)
   cg_models.append(cg)
   return CoveragePlan(covergroups=cg_models)
```

■ **Why We Wrote This Code:** Translates requirements into measurable verification targets. Functional coverage ensures that tests actually exercise all protocol states, illegal values, and boundary conditions.

■■ **How It Works & Logic:** Constructs SystemVerilog covergroups containing explicit bins for valid ranges, illegal bins, transition bins (e.g., IDLE => WRITE => READ => IDLE), and cross-coverage between control and data buses.

■ **Inter-Module Dataflow:** Outputs coverage_model.sv and coverage_spec.json for integration into UVM subscribers.

■ File Path: verisight/plan/assertion_architect.py

System Role: Formal SystemVerilog Assertion (SVA) Synthesizer

```
Core Implementation Code:
  class AssertionArchitect:
   def generate_assertions(self, features: List[RequirementItem]) -> List[AssertionItem]:
   # Synthesizes 30+ formal SVA concurrent assertions
   sva_list = []
   for feat in features:
   if feat.has_temporal_constraint:
   sva_list.append(self._create_sva_property(feat, op="|->", guard="!rst_n"))
   return sva_list
```

■ **Why We Wrote This Code:** Bridges natural language timing requirements into executable formal checks. SVA assertions run continuously during simulation, immediately flagging protocol violations at the exact clock cycle they occur.

■■ **How It Works & Logic:** Extracts timing constraints from specifications (e.g., 'TREADY must assert within 4 cycles of TVALID') and generates formal SVA properties using overlapping (|->) or non-overlapping (|=>) implication operators with reset guards.

■ **Inter-Module Dataflow:** Outputs assertions.sv bindable to DUT module instances.

■ File Path: verisight/plan/coverage_ingest.py

System Role: Simulator UCDB & vDB Coverage Database Ingester

```
Core Implementation Code:
  class CoverageIngest:
   def ingest_coverage(self, coverage_path: str):
   if coverage_path.endswith(".ucdb") or coverage_path.endswith(".xml"):
   return self._parse_questa_ucdb(coverage_path)
   elif coverage_path.endswith(".vdb") or "vdb" in coverage_path:
   return self._parse_vcs_vdb(coverage_path)
   return self._parse_generic_report(coverage_path)
```

■ **Why We Wrote This Code:** Closes the loop between planned coverage and actual simulation results. Chip teams use different commercial simulators; this module provides a unified ingestion layer for coverage databases.

■■ **How It Works & Logic:** Parses XML/text exports from Siemens Questa UCDB and Synopsys VCS vDB databases, extracting covergroup hierarchies and hit count statistics for every defined bin.

■ **Inter-Module Dataflow:** Feeds actual simulator coverage data to coverage_correlator.py.

■ File Path: verisight/plan/coverage_correlator.py

System Role: Dynamic Coverage Hole Analyzer & Directed Stimulus Generator

```
Core Implementation Code:
  class CoverageCorrelator:
   def find_holes_and_synthesize_stimulus(self, planned_model, actual_coverage):
   unhit_bins = self._extract_zero_hit_bins(planned_model, actual_coverage)
   directed_sequences = []
   for hole in unhit_bins:
   seq = self._synthesize_directed_uvm_sequence(hole)
   directed_sequences.append(seq)
   return CoverageClosureReport(unhit=unhit_bins, directed_sequences=directed_sequences)
```

■ **Why We Wrote This Code:** Eliminates the multi-week 'coverage closure stall'. Teams often get stuck at 85% coverage because random seeds cannot hit rare corner-case states. This code automatically creates targeted stimulus for unhit bins.

■■ **How It Works & Logic:** Subtracts actual hit bins from the planned coverage matrix to isolate unreached corner cases (e.g., concurrent read/write during FIFO almost-full), then generates constrained-random UVM sequences to hit them in days.

■ **Inter-Module Dataflow:** Outputs directed_sequences.sv and coverage_closure.md.

■ File Path: verisight/plan/traceability.py

System Role: ISO 26262 ASIL-D & DO-254 Traceability Matrix Exporter

```
Core Implementation Code:
  class TraceabilityMatrix:
   def build_matrix(self, features, tests, coverage, assertions):
   matrix_rows = []
   for f in features:
   row = {
   "req_id": f.id,
   "text": f.description,
   "tests": [t.id for t in tests if f.id in t.req_refs],
   "coverage": [c.name for c in coverage if f.id in c.req_refs],
   "sva": [a.name for a in assertions if f.id in a.req_refs]
   }
   matrix_rows.append(row)
   return self._export_csv_and_html(matrix_rows)
```

■ **Why We Wrote This Code:** Guarantees compliance for automotive and safety-critical chips. Standards like ISO 26262 ASIL-D and DO-254 require proving that 100% of functional requirements have corresponding tests, assertions, and coverage points.

■■ **How It Works & Logic:** Constructs a bidirectional requirement-to-verification matrix and exports audit-ready CSV (traceability_matrix.csv) and HTML reports, providing proof of verification completeness for safety audits.

■ **Inter-Module Dataflow:** Generates compliance deliverables in output/verification_plan/.

4. Engine 3 Deep Dive: Spec-Driven UVM 1.2 Synthesis

■ File Path: verisight/uvm/uvm_architect.py

System Role: Complete 14-Component UVM 1.2 VIP Environment Synthesizer

```
Core Implementation Code:
  class UVMArchitect:
   VIP_FILES = [
   "_if.sv", "_seq_item.sv", "_sequencer.sv", "_driver.sv", "_monitor.sv",
   "_agent.sv", "_reference_model.sv", "_scoreboard.sv", "_coverage.sv",
   "_sequence.sv", "_env.sv", "_test.sv", "_pkg.sv", "_tb_top.sv"
   ]
   def synthesize_suite(self, spec_truth, rtl_ports, output_dir):
   for f_type in self.VIP_FILES:
   code = self._generate_component(f_type, spec_truth, rtl_ports)
   self._write_and_validate(output_dir, f_type, code)
```

■ **Why We Wrote This Code:** Automates the most labor-intensive task in silicon verification. Writing a production UVM 1.2 testbench takes senior engineers 3 to 4 weeks. This code synthesizes a complete, standards-compliant 14-file UVM environment in seconds.

■■ **How It Works & Logic:** Generates all 14 modular components. Critically, it constructs an independent behavioral Golden Reference Model (_reference_model.sv) and Scoreboard (_scoreboard.sv) connected via UVM analysis FIFOs to discover RTL bugs independently.

■ **Inter-Module Dataflow:** Outputs 14 SystemVerilog UVM files to output/spec_driven_uvm/ or output/uvm_environment/.

■ File Path: verisight/uvm/matrix_synthesizer.py

System Role: Protocol Matrix & Transaction Decomposer

Core Implementation Code:

```
  class MatrixSynthesizer:
   def derive_sequence_item(self, protocol_info, features):
   rand_fields = []
   constraints = []
   for field in protocol_info.payload_fields:
   rand_fields.append(f"rand logic [{field.width-1}:0] {field.name};")
   constraints.append(f"constraint c_{field.name} {{ {field.name} inside {[field.legal_ranges]}; }}")
   return self._format_seq_item(rand_fields, constraints)
```

■ **Why We Wrote This Code:** Translates protocol specification parameters into synthesizable SystemVerilog transaction objects. Manually writing sequence items with proper rand variables and distribution constraints is tedious and prone to typos.

■■ **How It Works & Logic:** Decomposes protocol timing parameters and address boundaries into SystemVerilog sequence item fields, generating pre/post_randomize hooks and UVM automation macros (`uvm_field_int).

■ **Inter-Module Dataflow:** Used by uvm_architect.py to build _seq_item.sv and _driver.sv.

5. IEEE-1800 AST Compilers, Parsers & Waveform Slicing

■ File Path: verisight/parsers/pyslang_rtl.py

System Role: SystemVerilog IEEE-1800 Compiler & AST Elaborator

Core Implementation Code:

```
  def _build_always_block(proc: Any, sm: Any) -> Optional[AlwaysBlock]:
   pk = str(getattr(proc, "procedureKind", "")).split(".")[-1] # AlwaysFF, AlwaysComb
   events = _timing_events(proc) # Extract (edge, signal) e.g. (PosEdge, clk)
   written, read, nb_flags = _collect_assignments(proc)
   # Check if sequential block has a valid reset branch:
   has_reset = bool(reset) or bool(_RESET_IF_RE.search(raw_text))
   if pk == "AlwaysFF" and not has_reset:
   flag_uninitialized_flop_candidate(written)
```

■ **Why We Wrote This Code:** Solves the limitation of text-based AI. Generic LLMs cannot understand hardware semantics. We wrote this code to interface directly with Slang, the fastest modern C++ SystemVerilog compiler frontend.

■■ **How It Works & Logic:** Elaborates the real AST, extracts module port directions and bitwidths, identifies clock and reset sensitivity lists, and statically flags registers written in sequential blocks that lack a reset branch—pinpointing uninitialized 'X' bugs without simulation.

■ **Inter-Module Dataflow:** Provides compiler AST models to Agent 1, Agent 3, and the X-Tracer tool.

■ File Path: verisight/parsers/vcd_parser.py

System Role: Rust-Backed Temporal Waveform Slicer (+/- 20 Clock Cycles)

```
Core Implementation Code:
  class VCDParser:
   def slice_window(self, vcd_path: str, fail_timestamp: int, window_cycles: int = 20):
   # Uses pywellen / vcdvcd to extract only +/- 20 cycles around failure
   start_time = max(0, fail_timestamp - (window_cycles * clock_period))
   end_time = fail_timestamp + (window_cycles * clock_period)
   return self._extract_transitions(vcd_path, start_time, end_time)
```

■ **Why We Wrote This Code:** Solves the 50GB waveform memory explosion. Real SoC simulation dumps are tens of gigabytes, which crash Python processes and cannot fit into LLM context windows.

■■ **How It Works & Logic:** Uses fast C++/Rust bindings (pywellen) to perform targeted temporal slicing: it reads only the 20 clock cycles immediately surrounding the assertion failure timestamp, reducing gigabytes of trace data into a sub-megabyte JSON slice.

■ **Inter-Module Dataflow:** Provides temporal signal state transitions to Agent 1, Agent 3, and the Waveform Studio.

■ File Path: verisight/parsers/log_parser.py

System Role: Simulation Log Tokenizer & Error Pattern Classifier

```
Core Implementation Code:
  class LogParser:
   ERROR_PATTERNS = [
   re.compile(r"UVM_ERROR\s+([A-Z0-9_\/.]+)\((\d+)\)\s+@\s+(\d+)\s*:\s*(.*)"),
   re.compile(r"\*E,([A-Z0-9]+)\s+\(([^)]+),(\d+)\):\s*(.*)"), # Cadence Xcelium
   re.compile(r"%\s*Error:\s*(.*)") # Verilator
   ]
   def tokenize(self, log_path: str) -> LogData:
   # Extracts timestamps, source locations, and error messages
   ...
```

■ **Why We Wrote This Code:** Normalizes incompatible simulator outputs. Synopsys VCS, Siemens Questa, Cadence Xcelium, and Verilator all print errors in completely different formats. This code unifies them into a single standard schema.

■■ **How It Works & Logic:** Scans raw simulation logs with pre-compiled regex tokenizers, extracting assertion failure timestamps, scoreboard mismatch payloads, and file:line locations.

■ **Inter-Module Dataflow:** Feeds normalized error signatures to Agent 1 and Agent 2.

6. Hardware Execution, Simulation Sandbox & Tools

■ File Path: verisight/tools/sim_runner.py

System Role: Closed-Loop Multi-Simulator Sandbox Runner

Core Implementation Code:

```
  class SimRunner:
   def run_sandbox_regression(self, patch_diff: str, base_dir: Path) -> bool:
   # Create isolated temporary workspace in /tmp
   sandbox_dir = Path(tempfile.mkdtemp(prefix="verisight_sandbox_"))
   shutil.copytree(base_dir, sandbox_dir / "src")
   # Apply proposed patch inside sandbox
   self._apply_patch(sandbox_dir / "src", patch_diff)
   # Execute simulator (Verilator, Icarus, or VCS)
   success = self._run_simulator(sandbox_dir / "src")
   shutil.rmtree(sandbox_dir)
   return success
```

■ **Why We Wrote This Code:** Enforces the ultimate safety guardrail. Never allow an AI to commit code directly to a hardware repository. If an AI generates an invalid patch, this sandbox catches it before human engineers ever see it.

■■ **How It Works & Logic:** Clones the design repository into a temporary /tmp workspace, applies the candidate git patch, runs compilation and regression simulation, and returns True only if all tests pass with zero errors.

■ **Inter-Module Dataflow:** Used by Agent 5 during automated fix validation.

■ File Path: verisight/tools/sv_validate.py

System Role: Static IEEE-1800 Compiler Syntax Pre-Check Gateway

Core Implementation Code:

```
  class SVValidator:
   def validate_syntax(self, code_str: str) -> Tuple[bool, List[str]]:
   # Pre-compiles code through pyslang before writing to disk
   tree = pyslang.SyntaxTree.fromText(code_str)
   diags = [str(d) for d in tree.diagnostics if d.severity == pyslang.DiagnosticSeverity.Error]
   return (len(diags) == 0, diags)
```

■ **Why We Wrote This Code:** Prevents syntax hallucinations. An AI might generate a fix or testbench that misses a semicolon, misuses a keyword, or violates SystemVerilog grammar. This code acts as a compiler gatekeeper.

■■ **How It Works & Logic:** Feeds generated SystemVerilog code through pyslang's syntax tree parser. If diagnostics report errors, it returns the compiler diagnostic messages back to the agent so it can self-correct the code.

■ **Inter-Module Dataflow:** Invoked by Agent 5 and Engine 3 before saving any file.

■ File Path: verisight/tools/xtracer_runner.py

System Role: Static and Runtime X-Propagation Tracer Driver

Core Implementation Code:

```
class XTracerRunner:
 def find_uninitialized_regs(self, ast_tree) -> List[str]:
 # Traverses AST always_ff nodes to find registers missing reset logic
 uninitialized = []
 for block in ast_tree.always_blocks:
 if block.is_sequential and not block.has_reset:
 uninitialized.extend(block.assigned_signals)
 return uninitialized
```

■ **Why We Wrote This Code:** Isolates uninitialized flip-flop bugs that cause 4-state 'X' logic propagation. Tracing 'X' values manually in waveforms is one of the most tedious tasks in verification.

■■ **How It Works & Logic:** Scans sequential AST blocks for missing reset branches and traces driving gates backward from the failing cycle to pinpoint the uninitialized flop.

■ **Inter-Module Dataflow:** Used by Agent 3 during RTL root-cause analysis.

■ File Path: verisight/tools/eda_playground_exporter.py

System Role: Cloud EDA Playground 1-Click Exporter

Core Implementation Code:

```
class EDAPlaygroundExporter:
 def export(self, tb_dir: Path, rtl_dir: Path) -> str:
 # Bundles testbench, RTL, and run scripts into a single cloud package
 bundle = self._create_bundle(tb_dir, rtl_dir)
 share_url = self._upload_to_edaplayground(bundle)
 return share_url
```

■ **Why We Wrote This Code:** Facilitates collaboration. Remote team members or clients may not have local VCS or Questa licenses installed. This allows engineers to share a working simulation with one click.

■■ **How It Works & Logic:** Packages synthesized UVM testbenches, RTL source files, and run scripts into a clean bundle and generates a direct link to edaplayground.com.

■ **Inter-Module Dataflow:** Invoked by `./start edaplayground`.

7. Live Web Studio, FastAPI Server & Waveform Telemetry

■ File Path: verisight/server/app.py

System Role: FastAPI Asynchronous Web Studio & WebSocket Bridge

```
Core Implementation Code:
  app = FastAPI(title="VeriSight Web Studio")
  @app.websocket("/ws/simulate")
  async def websocket_simulation_stream(websocket: WebSocket):
   await websocket.accept()
   async for log_line in stream_simulator_output():
   await websocket.send_json({"type": "log_line", "data": log_line})
```

■ **Why We Wrote This Code:** Provides an interactive, sub-millisecond web interface for verification engineers. Enables live regression monitoring and dynamic coverage tracking in a modern browser UI.

■■ **How It Works & Logic:** Launches an asynchronous FastAPI server on port 8000 (via `./start serve`), streaming real-time simulation stdout over WebSockets and serving the Sage/Burgundy single-page dashboard.

■ **Inter-Module Dataflow:** Backend server powering verisight/server/static/app.js.

■ File Path: verisight/reporter/waveform_studio.py

System Role: Sub-Cycle Waveform Cursor Synchronizer

```
Core Implementation Code:
  class WaveformStudio:
   def synchronize_cursor(self, assertion_fail_time_ns: int):
   # Automatically positions inspection cursor at exact assertion timestamp
   return {
   "cursor_time": assertion_fail_time_ns,
   "highlight_signals": ["clk", "rst_n", "tvld", "trdy", "state"],
   "radix": "hex"
   }
```

■ **Why We Wrote This Code:** Eliminates manual waveform searching. In tools like Verdi, an engineer must manually copy an assertion timestamp from a log file and search for it in a giant waveform viewer. This code automates that cursor alignment.

■■ **How It Works & Logic:** Generates interactive canvas visualizers that automatically jump the inspection cursor to the exact nanosecond where the assertion tripped, decoding bus signals in binary, hex, and FSM state names.

■ **Inter-Module Dataflow:** Embedded into report.html and the Live Web Studio.

8. Enterprise Security, Sovereign Air-Gap & Memory

■ File Path: verisight/utils/airgap.py

System Role: Sovereign Air-Gapped Network Enforcer & Local Model Gateway

```
Core Implementation Code:
  class AirGapManager:
   def enforce_airgap(self):
   # Disables external socket connections; validates local Ollama / vLLM
   os.environ["HTTP_PROXY"] = ""
   os.environ["HTTPS_PROXY"] = ""
   assert self._probe_localhost_endpoint("http://localhost:11434")
```

■ **Why We Wrote This Code:** Guarantees that proprietary silicon IP never leaks to public cloud servers. Defense, aerospace, and semiconductor fabless companies require strict air-gapped on-premise execution.

■■ **How It Works & Logic:** Blocks external network requests, redirects LLM calls to local Ollama or vLLM server instances running open-weights models (DeepSeek-Coder-V2, Llama-3-70B), and verifies zero outbound packet transmission.

■ **Inter-Module Dataflow:** Activated by the `--airgap` flag or wizard Option 1.

■ File Path: verisight/utils/anonymizer.py

System Role: Proprietary RTL & Specification Identifier Scrambler

```
Core Implementation Code:
  class Anonymizer:
   def anonymize_code(self, code_str: str) -> Tuple[str, Dict[str, str]]:
   # Scrambles proprietary signal names: mem_array -> sig_001, secret_key -> sig_002
   anonymized = code_str
   mapping = {}
   for ident in extract_identifiers(code_str):
   fake_name = f"sig_{uuid.uuid4().hex[:6]}"
   mapping[fake_name] = ident
   anonymized = anonymized.replace(ident, fake_name)
   return (anonymized, mapping)
```

■ **Why We Wrote This Code:** Enables safe cloud LLM usage. If a customer chooses to use Google Gemini or Claude for maximum reasoning power, this code scrubs all proprietary names before the prompt leaves the machine.

■■ **How It Works & Logic:** Replaces module names, signal identifiers, and comments with randomized tokens, sends the scrubbed code to the cloud API, and then maps the original names back locally into the response.

■ **Inter-Module Dataflow:** Activated when cloud APIs are selected with anonymization enabled.

■ File Path: verisight/rag/knowledge_base.py

System Role: ChromaDB 11-Collection Vector Memory Store

```
Core Implementation Code:
  class KnowledgeBase:
   def __init__(self):
   self.client = chromadb.PersistentClient(path=".chroma_db")
   # 11 specialized collections: specs, registers, protocol_rules, past_bugs, etc.
   self.collections = [self.client.get_or_create_collection(c) for c in COLLECTIONS]
```

■ **Why We Wrote This Code:** Acts as institutional memory for the verification team. When an engineer debugs a tricky FIFO bug on one block, other engineers shouldn't have to re-solve the same bug 3 months later on a different block.

■■ **How It Works & Logic:** Stores tokenized failure signatures, specification paragraphs, and verified patches in 11 persistent vector collections, using semantic similarity search to instantly recall past solutions.

■ **Inter-Module Dataflow:** Queried by Agent 1 and Agent 2 during triage.